

Self-Assessment: Istanbul Retail Sales Dataset

Dataset: istanbul_sales_data.csv (503 transactions)

Instructions: Use Python (pandas, Matplotlib, Seaborn) in Spyder to complete the tasks. Document your process and include code, tables, and charts in your report.

Question 1 – Data Cleaning & Preparation (20 marks)

1. Check for missing values, duplicates, or zero quantities/prices.
 2. Remove or handle any problematic rows.
 3. Create a new column `total_revenue = quantity * price`.
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Question 2 – Exploratory Data Analysis (30 marks)

1. **Frequency tables:**
 - Product category (category)
 - Payment method (payment_method)
 - Customer gender (gender)
 2. **Age groups:**
 - Create age groups: 0-20, 21-30, 31-40, 41-50, 51-60, 61-70, 71+.
 - Calculate **total revenue per age group**.
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Question 3 – Visualization (30 marks)

1. Bar chart: **Total revenue per shopping mall**.
2. Pie chart: **Payment method distribution**.
3. Line chart: **Total revenue over time** (monthly or daily).
4. Histogram: **Distribution of customer ages**.
5. Scatter plot: **Customer age vs. total revenue**.

Bonus: Label your charts clearly with titles, axes, and colors.

Question 4 – Insights & Analysis (20 marks)

1. Identify the **top 3 revenue-generating categories**.
 2. Identify the **most valuable age group** (highest total spend).
 3. Identify the **most commonly used payment method**.
 4. Summarize any trends over time: e.g., which months have higher sales, or which malls perform best.
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Deliverables

- Python code in .py file or Jupyter Notebook.
 - Charts exported as images.
 - A short report (1–2 pages) explaining your **findings and insights**.
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